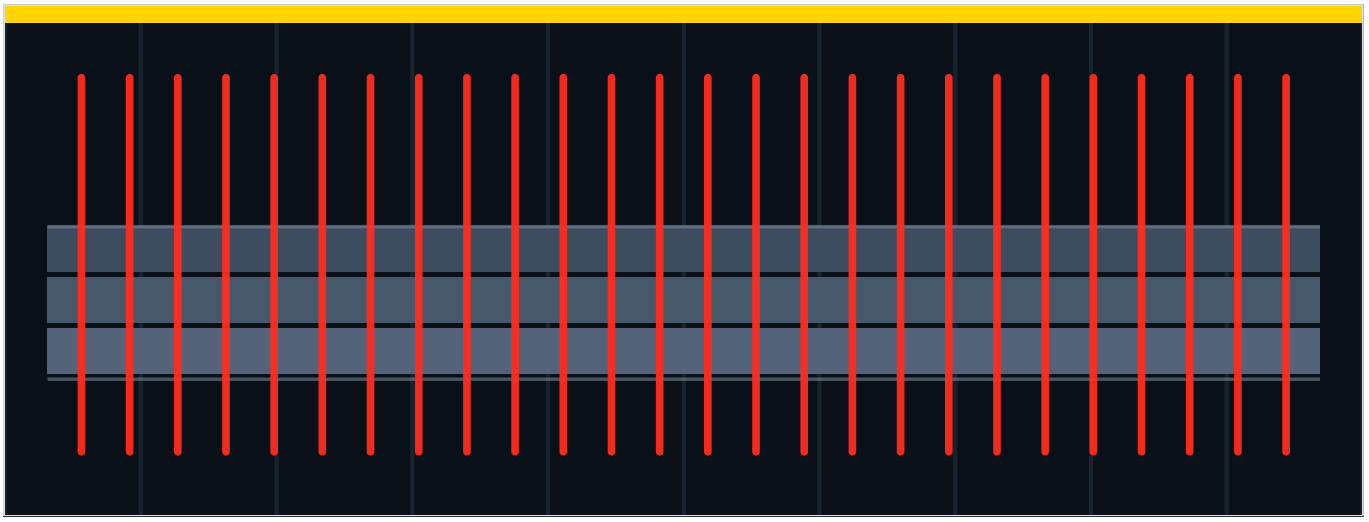
APPLICATION · HYGIENE PRODUCTS AND NONWOVEN

MANY LASER REFERENCE LINES ONE SERVO AXIS

In multi-layer nonwoven lines, a single recipe means dozens of projected cutting references. Measuring and adjusting each one individually turns the changeover into the longest stop of the shift. ServoMotion moves each carriage to its recipe position in sequence, and once setup is complete, all lines remain visible at the same time.



Shuttle	26 lasers	2,300 mm
SYSTEM USED IN THE EXAMPLE INSTALLATION	CARRIAGE COUNT	TRAVEL

ON THE LINE — DO YOU RECOGNISE THESE SYMPTOMS?

Setup grows with the number of projected laser lines

Measuring 26 blade positions one by one stretches the changeover well beyond the shift plan.

Small offsets accumulate

A millimetre of deviation on each slit adds up and becomes visible on the outermost cut.

Web tension hides the cause

When edge quality drops, it is not obvious whether the cause is the blade, the tension or the position.

No verification step

Setup errors only come to light on the first product, when the cost has already been incurred.

WHAT WE MEASURE DURING THE SITE SURVEY

Number of simultaneous reference lines	The blade count in your densest recipe determines the number of carriages.
Slit width range	The narrowest slit determines line thickness and positioning tolerance.
Total working width	The widest recipe on the line determines travel range and rail length.
Changeover frequency	The number of changeovers per day forms the basis of the payback calculation.
Rail and traverse structure	The rail plane translates directly into positional accuracy.
Automation interface	Whether recipe data can be read from the line controller defines the integration scope.

WHICH ARCHITECTURE FITS THIS SEGMENT?

MOST APPLICATIONS

SHUTTLE PRECISOR

Multi-layer nonwoven lines need dozens of reference lines visible at once. A single servo axis carries the laser carriages to their recipe positions in sequence; each carriage locks on the rail, and once setup is complete, all lines remain visible together.

IF THE NUMBER OF PROJECTED LASER LINES IS LOW

CORE PRECISOR

If only a few reference lines need to be visible at the same time, an enclosed industrial profile may be sufficient at a lower investment. The decision is based on the densest recipe.

EXAMPLE INSTALLATION

SHUTTLE PRECISOR · 26 LASERS · 2,300 MM TRAVEL

A three-rail shuttle architecture: one servo axis carries 26 carriages to their recipe positions in sequence, each carriage locks on its rail, and once setup is complete every line stays visible at once. These figures come from one example installation; the dimensions for your line are determined during the site survey.

This note summarises the typical requirements of the segment and one example installation. The final system architecture, housing size and scope of connection are established during the site survey. The lasers provide an alignment reference; material and machine elements are set by the operator. Values may change without notice as the products are developed.

LET US REVIEW YOUR LINE TOGETHER servomotion.com.tr/en/contact.html